BUSINESS ANALYTICS END-TERM ASSESSMENT

AIRBNB LISTINGS DATA ANALYSIS

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Introduction

This project analyzes Airbnb listings data to uncover the drivers of pricing and customer satisfaction. Using Exploratory Data Analysis (EDA), K-Means clustering, Logistic Regression, and Linear Regression, we evaluate structural property features and their impact on review scores.

The report is structured to provide:

- Comprehensive EDA with descriptive statistics, missing value treatment, outlier detection, and visualizations.
- Clustering analysis to identify distinct market segments.
- Regression models to assess predictors of customer satisfaction.
- Business interpretations and strategic recommendations for Airbnb hosts.

Exploratory Data Analysis (EDA)

Data Quality

Dataset size: 37,056 rows × 15 columns.

Missing values:
Review_Score_Rating: 8,395
host_identity_verified & host_has_profile_pic: 86 each

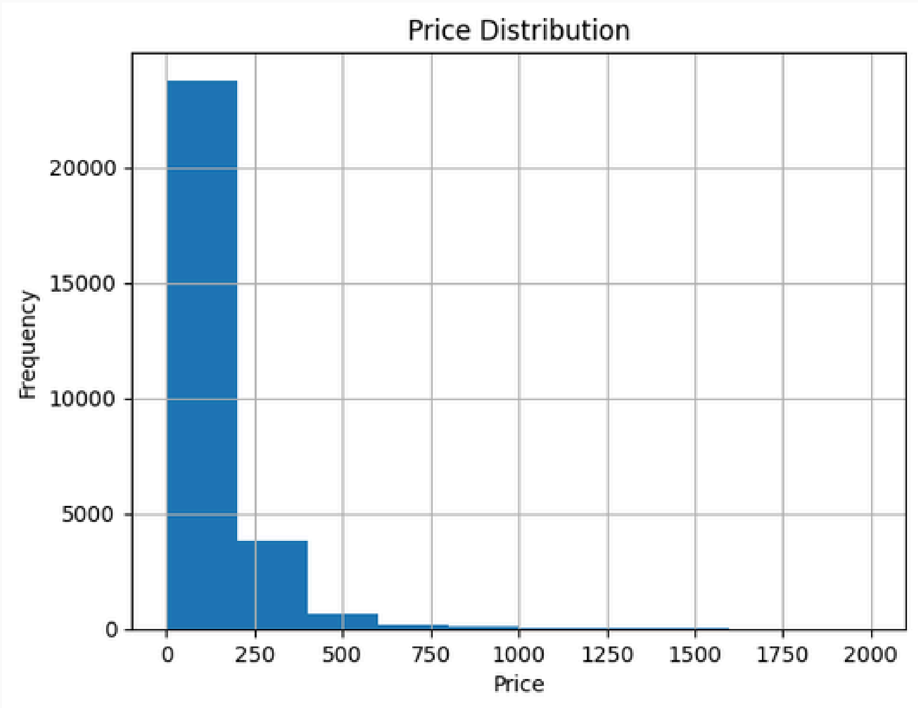
Treatment: Rows with missing critical numerical values were dropped, resulting in a cleaned dataset of 28,661 rows.

Data Description

	Price	Accommodates	Bathrooms	Number_of_Reviews	Bedrooms	Beds	Review_Score_Rating
count	28661.000000	28661.000000	28661.000000	28661.000000	28661.000000	28661.000000	28661.000000
mean	147.306863	3.213949	1.180664	27.086389	1.256411	1.732389	94.040054
std	134.170287	2.146648	0.523373	41.349821	0.838881	1.255462	7.925609
min	1.000000	1.000000	0.000000	1.000000	0.000000	0.000000	20.000000
25%	73.000000	2.000000	1.000000	3.000000	1.000000	1.000000	92.000000
50%	110.000000	2.000000	1.000000	11.000000	1.000000	1.000000	96.000000
75%	175.000000	4.000000	1.000000	33.000000	1.000000	2.000000	100.000000
max	1999.000001	16.000000	8.000000	605.000000	10.000000	16.000000	100.000000

Graphs & Interpretations

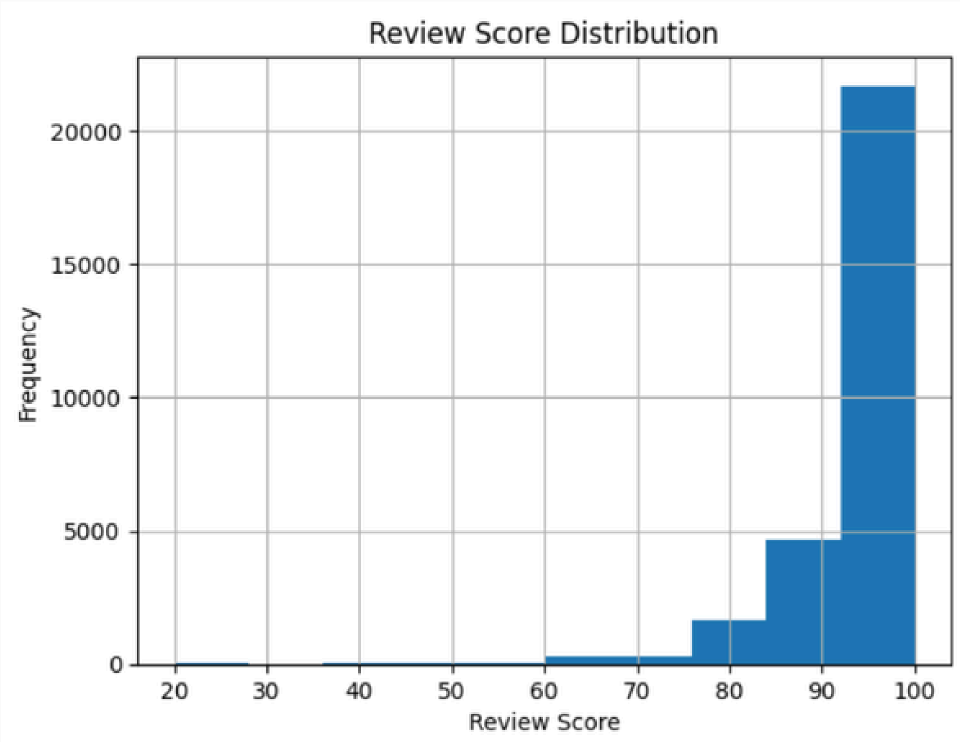
Price Distribution Histogram



Interpretation:
The histogram of Price shows a right-skewed distribution. Most listings fall between 50–200 USD, which indicates that Airbnb's market is dominated by budget and mid-range properties. Outliers above 1000 USD represent luxury properties that cater to niche segments.

Business Insight:
Hosts targeting mainstream customers should price competitively in the 50–200 USD range, while luxury hosts should differentiate through premium services.

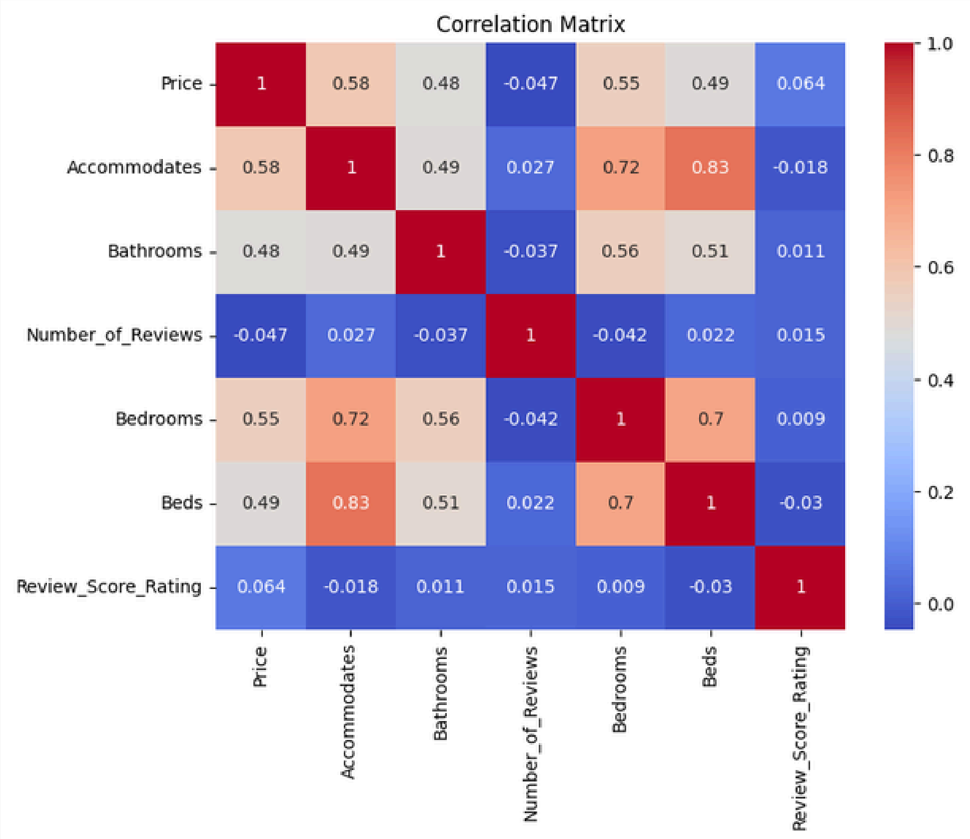
Review Score Distribution Histogram



Interpretation:
The histogram of Review_Score_Rating reveals that the majority of properties are rated between 90–100. Very few listings fall below 80, suggesting that customers generally provide high ratings.

Business Insight:
Airbnb’s customer base tends to rate generously, which implies that maintaining a rating above 95 is achievable with consistent service quality. Hosts should focus on small service improvements to push ratings into the “very high” category (≥98).

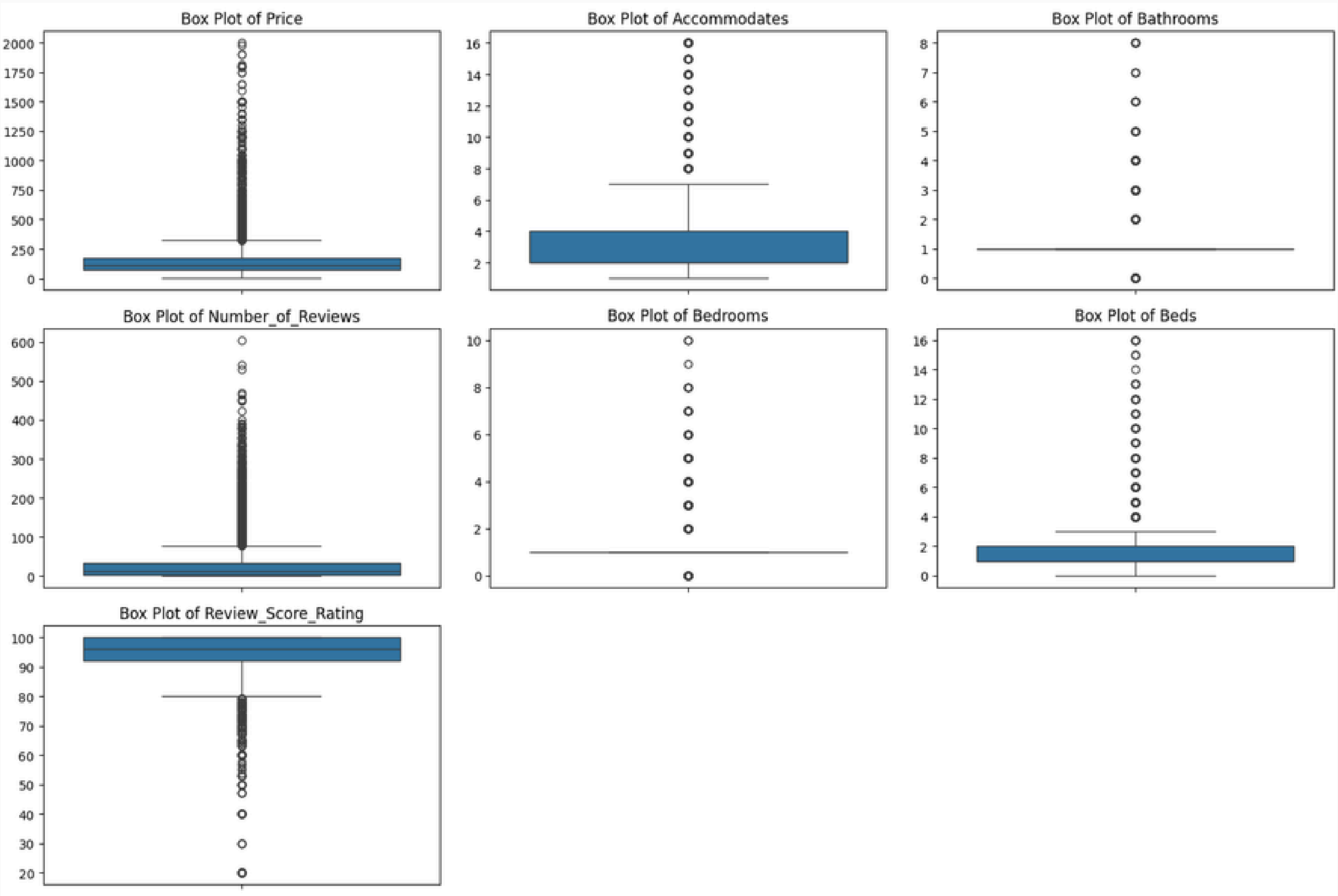
Correlation Between Price and Property Characteristics



Interpretation:
The heatmap shows correlations among numerical variables:
Price is strongly correlated with **Accommodates (0.58), Bedrooms (0.55), and Beds (0.49)**.
Review scores show negligible correlation with structural features.

Business Insight:
Larger properties are priced higher, but customer satisfaction is not structurally linked. This suggests that service quality and host behavior matter more than property size or price.

Boxplots for Outlier Detection



Boxplots for Outlier Detection

Price:

- Median around 100–150 USD.
- Outliers above 1000 USD represent luxury properties.
- Business Insight: Luxury listings are rare and need differentiated marketing.

Number of Reviews:

- Most listings have fewer than 50 reviews.
- Outliers above 500 reviews indicate long-standing, popular properties.
- Business Insight: Highly reviewed properties can serve as benchmarks.

Accommodates:

- Majority accommodate 2–4 guests.
- Outliers above 10 guests represent large group accommodations.
- Business Insight: Large listings cater to niche family/group travel segments.

Bedrooms:

- Most listings have 1–2 bedrooms.
- Outliers with 8–10 bedrooms represent rare luxury villas.
- Business Insight: Villas are unique offerings, not mainstream.

Beds:

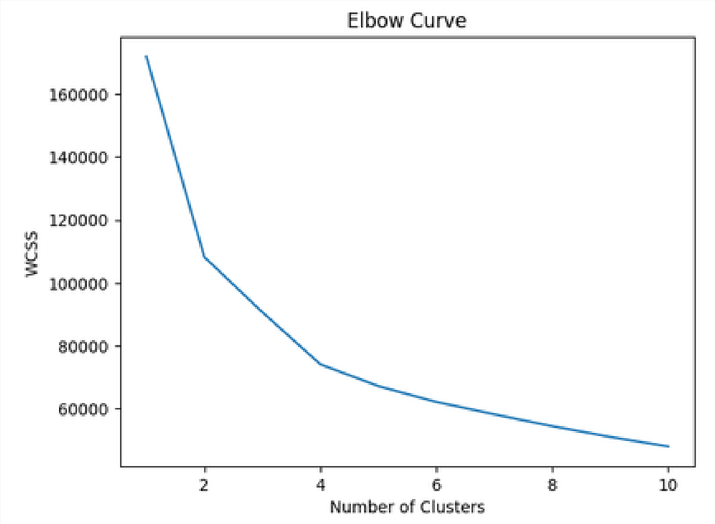
- Most listings have 1–3 beds.
- Outliers with 12–16 beds suggest dorm-style or group accommodations.
- Business Insight: Overcrowding may reduce satisfaction.

Bathrooms:

- Most listings have 1–2 bathrooms.
- Outliers with 6–8 bathrooms represent luxury properties.
- Business Insight: More bathrooms don’t necessarily improve ratings, showing diminishing returns.

K-Means Clustering

Elbow Curve(Clustering preparation)



Interpretation:
The elbow curve shows a clear bend at k=3, suggesting that three clusters are optimal.

Business Insight:
Airbnb listings can be segmented into three distinct market categories: budget-friendly, premium, and affordable-high engagement. This segmentation helps hosts identify their competitive positioning.

Cluster Profiles

Cluster	Price	Accommodates	Bathrooms	Reviews	Bedrooms	Beds
0	1.159	25	105	155	10	136
1	3.404	71	196	220	275	393
2	1.194	30	106	1.268	10	155

Observations:
Cluster 0: Budget-friendly, small properties.
Cluster 1: Premium, large properties.
Cluster 2: Affordable properties with high customer engagement.

Business Insight:
Cluster 2 is the sweet spot — affordable listings with strong customer engagement.

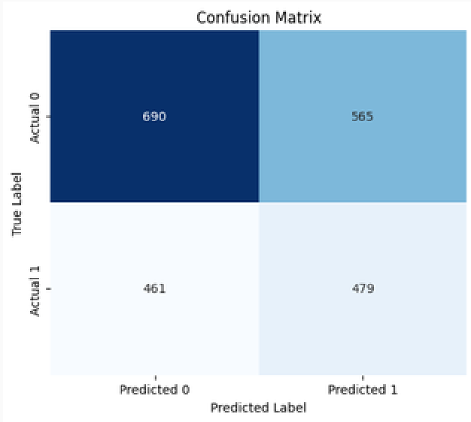
Logistic Regression

```
Accuracy Score: 0.5325740318906605
Precision Score: 0.45881226053639845
Recall Score: 0.5095744680851064

Classification Report:
              precision    recall  f1-score   support

           0       0.60      0.55      0.57      1255
           1       0.46      0.51      0.48       940

    accuracy                           0.53      2195
   macro avg       0.53      0.53      0.53      2195
weighted avg       0.54      0.53      0.53      2195
```




```
         Current function value: 0.677161
         Iterations 4
                           Logit Regression Results
==============================================================================
Dep. Variable:         Very_High_Score   No. Observations:                19748
Model:                           Logit   Df Residuals:                    19742
Method:                            MLE   Df Model:                            5
Date:                 Sat, 21 Feb 2026   Pseudo R-squ.:                0.009403
Time:                         07:21:29   Log-Likelihood:                -13373.
converged:                        True   LL-Null:                       -13500.
Covariance Type:             nonrobust   LLR p-value:                 8.062e-53
================================================================================
                   coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------
const           -0.1165      0.073     -1.589      0.112      -0.260       0.027
Price            0.0026      0.000     11.895      0.000       0.002       0.003
Bathrooms       -0.0403      0.054     -0.743      0.457      -0.146       0.066
Bedrooms         0.1253      0.032      3.881      0.000       0.062       0.189
Beds            -0.2804      0.027    -10.321      0.000      -0.334      -0.227
Cleaning_Fee    -0.2205      0.034     -6.397      0.000      -0.288      -0.153
================================================================================
```

The pseudo R^2 value of 0.009 indicates that the selected structural variables explain less than 1% of the variation in extremely high ratings. Although some predictors such as price, bedrooms, beds, and cleaning fee are statistically significant, their overall explanatory power is limited. This suggests that achieving very high ratings (≥98) is not primarily driven by physical property characteristics.

Linear Regression

```
R-squared Score: 0.0003778959525350123
                            OLS Regression Results
==============================================================================
Dep. Variable:     Review_Score_Rating   R-squared:                       0.011
Model:                             OLS   Adj. R-squared:                  0.010
Method:                  Least Squares   F-statistic:                     42.31
Date:                 Sat, 21 Feb 2026   Prob (F-statistic):           1.66e-43
Time:                         07:21:29   Log-Likelihood:                -69848.
No. Observations:                19748   AIC:                         1.397e+05
Df Residuals:                    19742   BIC:                         1.398e+05
Df Model:                            5
Covariance Type:             nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const           93.7802      0.299    313.384      0.000      93.194      94.367
Price            0.0098      0.001     11.161      0.000       0.008       0.011
Bathrooms       -0.5943      0.220     -2.697      0.007      -1.026      -0.162
Bedrooms         0.4186      0.131      3.190      0.001       0.161       0.676
Beds            -0.9276      0.107     -8.647      0.000      -1.138      -0.717
Cleaning_Fee     0.7449      0.142      5.252      0.000       0.467       1.023
==============================================================================
Omnibus:                    14656.321   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           338945.183
Skew:                          -3.374   Prob(JB):                         0.00
Kurtosis:                      22.141   Cond. No.                         805.
==============================================================================
```

The R^2 value of 0.011 indicates that only 1.1% of the variation in review scores is explained by the selected independent variables. While some predictors are statistically significant, their practical impact on overall rating is minimal. This implies that structural property features alone do not strongly determine customer satisfaction.

Coefficients & Interpretation
Price: Positive — higher prices slightly associated with higher ratings.
Bedrooms: Positive — more bedrooms improve ratings.
Beds: Negative — overcrowding reduces ratings.
Cleaning Fee: Positive — signals cleanliness.
Bathrooms: Negative — counterintuitive, may reflect data quirks.

Summary

- This project explored Airbnb listings data to identify the structural and behavioral factors influencing pricing and customer satisfaction. Using a dataset of over 37,000 entries, we applied exploratory data analysis (EDA), clustering, and regression modeling to uncover patterns and business insights.

- Key findings include:
  - Most listings are priced between 50–200 USD and rated between 90–100, indicating a generally satisfied customer base.
  - Outlier detection revealed niche luxury properties and highly reviewed listings that warrant separate strategic attention.
  - K-Means clustering identified three distinct market segments: budget-friendly, premium, and affordable-high engagement. Cluster 2 emerged as the most promising for hosts seeking high engagement at moderate pricing.
  - Logistic and linear regression models showed that structural features (price, bedrooms, beds, cleaning fee) are statistically significant but explain less than 1% of variation in review scores.
  - Customer satisfaction is not primarily driven by property attributes but by qualitative factors such as host responsiveness, service quality, and location.

Conclusion

- The analysis confirms that while structural features like price and size influence listing characteristics, they do not meaningfully predict customer satisfaction. High ratings on Airbnb are more strongly associated with intangible elements — communication, cleanliness, and overall guest experience.
- For hosts, this means that success lies not in maximizing physical features, but in delivering consistent, personalized, and high-quality service. For Airbnb as a platform, the findings suggest that host training, guest feedback analysis, and service quality metrics should be prioritized to enhance user satisfaction.
- Future research should incorporate qualitative data (e.g., review text, host behavior) and location-based variables to build more predictive models. By combining structured and unstructured data, Airbnb and its hosts can better understand what truly drives customer loyalty and long-term success.